Project Proposal

Web3 Smart Contract Security Scanner

1. Project Title

Web3 Smart Contract Security Scanner and Vulnerability Analysis Dashboard

2. Introduction

With the rapid growth of **blockchain and Web3 applications**, smart contracts have become a core component of decentralized systems. However, poorly written or insecure smart contracts have resulted in **millions of dollars in financial losses** due to vulnerabilities such as re-entrancy, improper access control, and unchecked external calls.

This project aims to design and develop a **security-focused Web3 tool** that analyses Solidity smart contracts, detects common vulnerabilities, and presents the results through a **user-friendly web dashboard**. The system helps developers and security researchers identify security issues early, improving the overall security of blockchain-based applications.

3. Problem Statement

Many developers deploy smart contracts without thorough security testing. Existing professional tools are either **expensive**, **complex**, or not beginner-friendly. As a result, small teams and learners often ignore security testing, leading to exploitable contracts.

There is a need for a **simple, accessible, and security-oriented smart contract scanning tool** that can detect common vulnerabilities and clearly explain risks.

4. Objectives

The main objectives of this project are:

- To analyse Solidity smart contracts for common security vulnerabilities
- To classify detected issues based on **risk level (Low, Medium, High)**
- To provide clear explanations of each vulnerability
- To present results through a **web-based dashboard**
- To promote secure smart contract development practices

5. Scope of the Project

The system will allow users to:

- Upload or paste Solidity smart contract code
- Automatically scan the code using static analysis techniques
- Detect common smart contract vulnerabilities
- View a structured security report with explanations and risk levels

In-Scope Vulnerabilities:

- Re-entrancy attacks
- Integer overflow / underflow
- Unchecked external calls
- Improper access control
- Hardcoded sensitive values

Out of Scope:

- Full blockchain deployment
- Live on-chain exploitation
- Advanced symbolic execution (future enhancement)

6. Tools & Technologies

Cybersecurity: Static code analysis

- Vulnerability pattern detection
- OWASP Smart Contract Top 10 concepts

Blockchain / Web3:

- Solidity
- Ethereum smart contract structure

Development Stack:

- Backend: **Python (Flask)** or **Node.js**
- Frontend: HTML, CSS, JavaScript
- Data format: JSON
- Version Control: Git & GitHub

7. System Workflow

1. User uploads or pastes a Solidity smart contract
2. Backend processes the contract code
3. Security engine scans for vulnerability patterns
4. Detected issues are categorized by risk level
5. Results are displayed on the web dashboard as a security report

8. Expected Outcomes

1. A functional Web3 smart contract security scanner
2. Clear vulnerability detection with explanations
3. A professional-looking security dashboard
4. Improved awareness of smart contract security risks
5. A strong portfolio project combining **Cybersecurity + Blockchain + Web3**

9. Future Enhancements

- Integration with live blockchain networks
- Support for multiple smart contract files
- PDF export of security reports
- AI-based vulnerability detection
- Wallet and dApp security analysis

10. Conclusion

The **Web3 Smart Contract Security Scanner** addresses a critical need in modern blockchain development by combining cybersecurity principles with Web3 technology. This project not only enhances smart contract security awareness but also serves as a practical, industry-level solution suitable for academic and professional use.